

WORCAP 2026 — Monthly precipitation forecast

Team: NoTime

Task and result

Predict monthly precipitation in mm/day on the competition grid for January 2023–December 2024. The participant reported Kaggle RMSE **1.53208** for this execution and **1.53551** for the original recipe. These scores were not measured locally; private score and rank remain pending.

Architecture and features

The five models are a spatial U-Net, a LightGBM residual corrector, two PoET temporal correctors with seeds 42 and 2026, and a LightGBM corrector using GEFS. Inputs combine competition observations with ECMWF SEAS5, DWD, Météo-France and NOAA GEFS forecasts.

The U-Net uses 25 channels: atmospheric lags, climatology, location, calendar, and SEAS5 precipitation anomaly and spread. The first LightGBM adds six- and twelve-month atmospheric summaries, precipitation references, the U-Net prediction and spatial descriptors. Its output joins SEAS5 atmospheric statistics and calibrated DWD/Météo-France forecasts in a 98-channel context. PoET attends to SEAS5, DWD and GEFS ensemble members and applies spatial convolutions. The final LightGBM adds GEFS weekly precipitation and coverage features.

The final blend assigns **0.6337257586287943** to the mean PoET prediction and **0.3662742413712057** to the GEFS corrector. Exact transformations and hyperparameters are defined in `src/worcap_forecast/` and its `config/training.json`.

Training and validation

Chronological prefix training produces 192 out-of-sample monthly contexts for 2007–2022, using 13 U-Net fits and nine first-stage LightGBM fits. These contexts train both PoET models and the GEFS corrector. Normalization and calibration use the available prefix; training labels end in December 2022. Each test prediction uses observations through its own previous month.

First-stage LightGBM OOF RMSE is **1.754724** for 2007–2022 and **1.751633** for 2015–2020. These values do not evaluate the final ensemble. Inference runs all five models and writes 1,885,464 rows in official sample order. Ten integrity, chronology and reconstruction tests passed.

Execution and data audit

Local execution requires Linux, Python 3.12, PyTorch 2.8.0/CUDA 12.8 and LightGBM 4.6.0. Recommended resources are 64 GiB RAM and a CUDA GPU with 16 GiB VRAM. OOF preparation took 17min37s, final training 8min18s and inference 32s. The [README](#) provides both prepared-artifact and complete reconstruction commands, with pinned environments and storage requirements.

Competition originals were imported and hash-verified; CDS and NOAA inputs were downloaded independently. Receipts bind data, source code, models and CSV. The [audit](#) records one unresolved evidence question: publication dates of historical hindcasts and reforecasts at simulated forecast origins. Copernicus public access was confirmed by an organizer; historical availability requires review. Data and weight redistribution terms are separate from the code's MIT license.